
Comparative Study of WGAN-GP and Lightweight Diffusion Models for Data Augmentation in Chest X-Ray Diagnosis

Junyi Xu

Jacobs School of Engineering
University of California San Diego
jux022@ucsd.edu

Ke Liu

Jacobs School of Engineering
University of California San Diego
ke1097@ucsd.edu

Abstract

Medical image diagnosis often suffers from significant class imbalance. Although public datasets may over-represent diseases, typical hospital data streams do not. This project proposes a comparative study of generative models to synthesize realistic Chest X-Rays for data augmentation. Specifically, we investigate the trade-off between computational efficiency and generation quality by comparing WGAN-GP against a Lightweight Denoising Diffusion Probabilistic Model (DDPM). We utilize the Kaggle Chest X-Ray Images (Pneumonia) dataset, downsampled to 64×64 to accommodate local hardware constraints (RTX 3060). Generated data will be evaluated using Fréchet Inception Distance (FID), and its clinical utility validated by the downstream performance improvement of a ResNet-18 classifier.

1 Research Problem

Automated diagnosis of pneumonia from Chest X-Rays suffers significantly from class imbalance. In real-world clinical datasets, positive cases are often scarce compared to healthy controls [1]. This data bias causes deep learning classifiers to bias towards the majority class [2], leading to high false-negative rates, namely missed diagnoses, which is clinically unacceptable [3].

While data augmentation is a standard solution, traditional methods such as rotation, flipping, *etc.* fail to introduce new pathological variations. Generative AI offers a promising alternative; however, training state-of-the-art models, like Stable Diffusion, is computationally prohibitive for standard clinical workstations. Therefore, this research aims to solve the problem of **generating high-quality, pathologically accurate synthetic X-rays under computational constraints** to effectively augment training datasets.

2 Significance and Impact

Clinical Utility: Improving the Recall of diagnostic algorithms directly enhances patient safety by ensuring early detection. A generative augmentation tool can significantly boost model performance in hospitals with limited data.

Accessibility: By focusing on lightweight architectures, this research demonstrates that effective medical AI augmentation can be performed on consumer-grade hardware.

Privacy-Preserving Data Sharing: Synthetic data offers a viable pathway for sharing high-quality medical data without violating privacy regulations (e.g., Health Insurance Portability and Accountability Act, HIPAA).

3 Related Work

Data augmentation based on Generative Adversarial Networks (GANs) has been widely explored. [4] demonstrated that DCGAN-generated synthetic images improved CNN performance in liver lesion classification. Madani *et al.* [5] applied GANs to chest X-rays, proving generative models can capture anatomical structures. However, standard GANs often suffer from Mode Collapse and training instability. Gulrajani *et al.* [6] proposed WGAN-GP (Wasserstein GAN with Gradient Penalty) to mitigate these issues, yet GANs still struggle with fine-grained texture diversity.

Denosing Diffusion Probabilistic Models (DDPMs) have shown superior image quality and mode coverage compared to GANs [7]. However, their slow sampling speed and high memory consumption remain significant barriers for practical deployment. This project aims to bridge this gap by optimizing Diffusion models for medical image synthesis on smaller scales.

4 Proposed Solution

4.1 Dataset and Preprocessing

We will utilize the **Kaggle Chest X-Ray Images (Pneumonia)** dataset. To ensure data authenticity while fitting local hardware constraints (NVIDIA RTX 3060), high-resolution images will be downsampled to 64×64 . This resolution preserves essential anatomical features (e.g., lung opacity, rib structure) without introducing interpolation artifacts associated with upsampling low-resolution datasets like PneumoniaMNIST [8].

4.2 Pipeline

Stage 1: Generative Quality Assessment

- **Input:** We will utilize the official *Training Set* (64×64 , normalized) to train WGAN-GP and Lightweight DDPM. We will try 128×128 if possible
- **Synthesis:** Once converged, each model generates a synthetic pathological dataset.
- **Metric:** Quality is evaluated using **Fréchet Inception Distance (FID)**. The original training set serves as ground truth.

Stage 2: Clinical Utility Validation

- **Setup:** A ResNet-18 classifier is trained under two conditions: (1) *Baseline* (real *Training Set* only) and (2) *Augmented* (real + synthetic data).
- **Testing:** Both classifiers are evaluated on the official, unseen *Test Set*.
- **Metrics:** Given class imbalance, we prioritize **AUC-ROC** and **F1-Score**. We also report **Recall** to ensure detection of critical cases is not compromised.

5 Novelty

Systematic Comparison: While GANs and Diffusion models exist separately in literature, this project provides a direct head-to-head comparison of their trade-offs between quality and efficiency, specifically for imbalanced medical datasets.

Hardware-Aware Optimization: We propose a "Mini-Diffusion" architecture, exploring the boundaries of high-quality generation on consumer-grade hardware.

6 Tentative Timeline

- **Week 1:** Data acquisition and preprocessing (resize the data to 64×64 and normalization).
- **Week 2-3:** Train WGAN-GP and Lightweight DDPM (64×64). Attempt 128×128 if possible.
- **Week 4:** Evaluation. Compute FID scores and run ResNet-18 experiments.
- **Week 5:** Final analysis, visualization, and report writing.

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